

Perceived Monetary Policy Rule: Identification and Macroeconomic Implications

Husang Kim ¹

¹Department of Economics
University of Texas at Austin

August 26, 2026

Motivation and Research Question

1. Taylor rule shows how actively monetary policy responds:

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Research Question

1. *What are the macroeconomic implications of the perceived rule ($\neq$ actual rule)?*
2. *How do we identify the perceived rule ($\phi_{\pi,t|t}$)?*

Overview - Identification

1. Identifying the bias in **perceived**-rule estimation

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- *Nowcast*: Real-time projection of the *current* period value.

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- *Taylor principle*: Raise i more than 1:1 with π .
- *Forecast-based*: Taylor principle **✗** ($\phi_{\pi,t|t} < 1$).
- *Nowcast-based*: Taylor principle **✓** ($\phi_{\pi,t|t} > 1$).

Overview - Macroeconomic Implications

Subjective Perception Markov Switching New Keynesian (SP-MSNK) model.

Perception ($\phi_{\pi,t|t}$) vs. Direct Policy ($\phi_{\pi,t}$)

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- Determinacy depends on whether agents **believe** in it, not the actual rule.
- **✗** Implement the Taylor principle **✓** Anchor **perception** to it.

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- ✗ The shift in monetary policy regime ✓ The shift in **perception**.
- Without it, peak inflation 4.8% → 7.4% and volatility $\times 1.93$.

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3. Quantifying the cost of misreading **perception** (2021-23)

- Optimal lift-off timing in 2021-2023?
- **N**-based (2021Q1) vs. **F**-based (2021Q3).

Contribution to the literature

1. **Macroeconomic effects of policy perceptions:** Bauer, Pflueger, and Sunderam 2024, Schorfheide 2005, Hamilton, Pruitt, and Borger 2011, Bocola et al. 2024. . .
⇒ Evaluate GE consequences for when perceived rules differ from actual rules
2. **Estimating perceived monetary policy rules:** Hamilton, Pruitt, and Borger 2011; Bundick 2015; Jia, Shen, and Zheng 2023; Bocola et al. 2024; Bauer, Pflueger, and Sunderam 2024. . .
⇒ Identify structural covariance bias and propose nowcast-based identification
3. **Determinacy and the Great Moderation:** Lubik and Schorfheide 2004, Coibion and Gorodnichenko 2011, Bianchi 2013, Sims and Zha 2006. . .
⇒ Reinterpret Great Moderation as driven by “Good Perceived Policy”
4. **Regime-switching DSGE models:** Davig and Leeper 2007; Bianchi 2013; Barthélemy and Marx 2017; Bianchi, Lettau, and Ludvigson 2022. . .
⇒ Relax FIRE assumption to isolate the perception channel.
5. **Post-pandemic policy and credibility:** Nakamura, Riblier, and Steinsson 2025, Bocola et al. 2025. . .
⇒ Show perceived Taylor principle expands the flexibility of the central bank.

Roadmap

1. **Simple Model of Perceived vs. Actual rule**
2. Identification of Perceived Rule
3. Empirical Estimation of Perceived Rule
4. Subjective Perception Markov Switching New Keynesian Model.

Perceived Rule vs. Actual Rule - Simple Model

Actual Model

$$i_t = E_t^P[\pi_{t+1}] + r_t$$

$$r_t = \rho_r r_{t-1} + v_t$$

$$i_t = \phi_f \pi_t$$

Subjective Model

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- $E_t^P[\cdot]$: private sector expectation under its **subjective** model.
- Agent observes r_t perfectly, but $\phi_p \neq \phi_f$.
- FIRE is the special case $\phi_p = \phi_f$, where $E_t^P[\cdot] = \mathbb{E}_t[\cdot]$.

Perceived Rule vs. Actual Rule - Simple Model

Perceived Law of Motion

$$i_{t+h|t} = \rho_r^h \underbrace{\frac{\phi_p}{\phi_p - \rho_r} r_t}_{i_{t|t}}$$

$$\pi_{t+h|t} = \rho_r^h \underbrace{\frac{1}{\phi_p - \rho_r} r_t}_{\pi_{t|t}}$$

Actual Law of Motion

$$i_t = \frac{\phi_p}{\phi_p - \rho_r} r_t$$

$$\pi_t = \underbrace{\frac{\phi_p}{\phi_f} \left[\frac{1}{\phi_p - \rho_r} r_t \right]}_{\text{FIRE } (\phi_p = \phi_f)} = \underbrace{\frac{1}{\phi_f} \left(\frac{\rho_r}{\phi_p - \rho_r} r_t \right)}_{E_t^P[\pi_{t+1}]} + \frac{1}{\phi_f} r_t$$

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Takeaways

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1. Determinacy: *perceived* Taylor principle ($\phi_p > 1$), NOT $\phi_f > 1$.
2. Perception Channel (through $E_t^P[\pi_{t+1}]$) vs. Direct Policy channel
3. Regressing $i_{t+h|t}$ on $\pi_{t+h|t}$ recovers ϕ_p , at any h .

Identification of Perceived Rule

Conventional practice regresses interest rate *forecasts* on inflation *forecasts*:
Hamilton, Pruitt, and Borger (2011) and Bauer, Pflueger, and Sunderam (2024)

$$E_t[i_{t+h}] = \beta_{OLS,IV} E_t[\pi_{t+h}] + e_{th} \quad \Rightarrow \quad \hat{\beta}_{OLS,IV} = \phi_p$$

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Takeaways

1. The covariance implies *downward* bias.
2. This does not depend on $E_t[\cdot] \neq \mathbb{E}_t[\cdot]$ or Perceived $\neq$ Actual.

Source of Covariance Bias - Illustration

Lubik and Schorfheide (2004) and Farmer, Waggoner, and Zha (2011):

$$x_t = \mathbb{E}_t x_{t+1} - \tau (i_t - \mathbb{E}_t \pi_{t+1})$$

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa x_t + z_{S,t},$$

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$$\phi_{\pi, s_t} \in \{\phi_{\pi, 1}, \phi_{\pi, 2}\}, \phi_{\pi, 1} > \phi_{\pi, 2} \text{ with } P = \begin{bmatrix} p & 1-p \\ 1-q & q \end{bmatrix}$$

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The solution of this model is

$$\pi_t = \begin{cases} a_{1,S} z_{S,t} & \text{if } s_t = 1 \\ a_{2,S} z_{S,t} & \text{if } s_t = 2 \end{cases}$$

Source and Sign of Covariance Bias

At time t , with $s_t = j \in \{1, 2\}$:

$$\text{COV} [\phi_{\pi, s_{t+h}}, \pi_{t+h} \mid s_t = j] = \underbrace{[P^h]_{j1} (1 - [P^h]_{j1}) (\phi_{\pi, 1} - \phi_{\pi, 2})}_{\text{Regime uncertainty (+)}} \times \underbrace{(a_{1,s} - a_{2,s})}_{\text{Rate feedback, GE (-)}} \rho_S^h z_{St}$$

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When is it nonzero?

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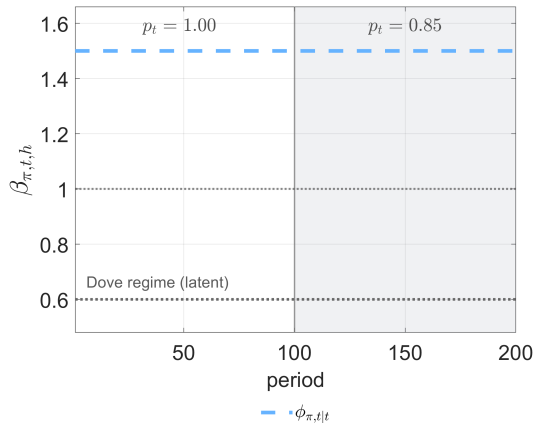
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Sign of Bias

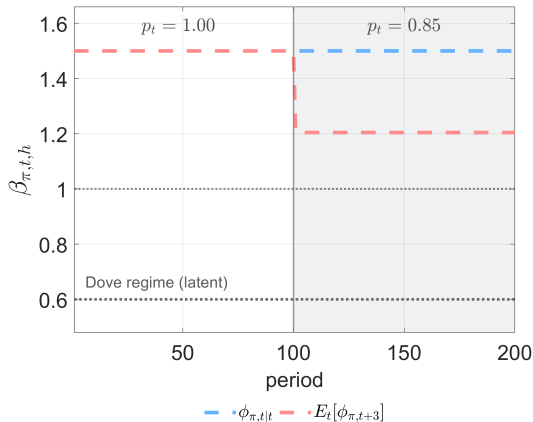
- $z_{St} > 0 \Rightarrow \mathbb{E}_t[\pi] > 0, \mathbb{C}_t < 0.$
- $z_{St} < 0 \Rightarrow \mathbb{E}_t[\pi] < 0, \mathbb{C}_t > 0.$

Numerical Illustration - Forecast vs. Nowcast



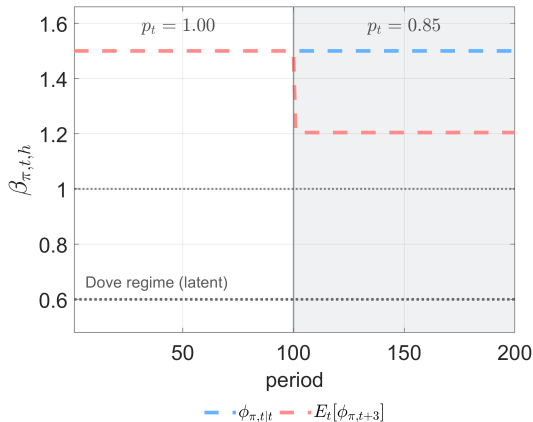
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- Always Hawk: $\phi_{\pi,t|t} = 1.50$.
- But starts to doubt: $\mathbb{E}_t[\phi_{\pi,t+3}] = 1.20$.

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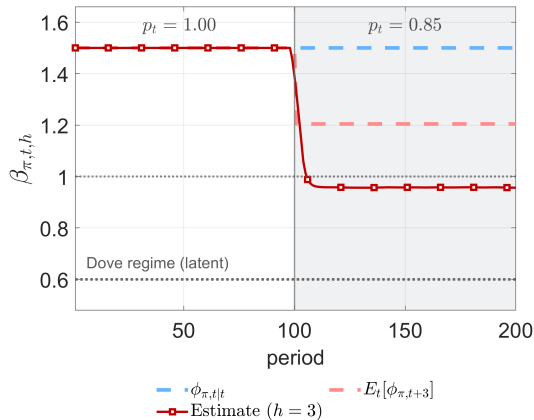


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TVP estimation:

$$E_t[i_{t+h}] = \beta_{\pi,t,h} E_t[\pi_{t+h}] + e_{t,h}$$

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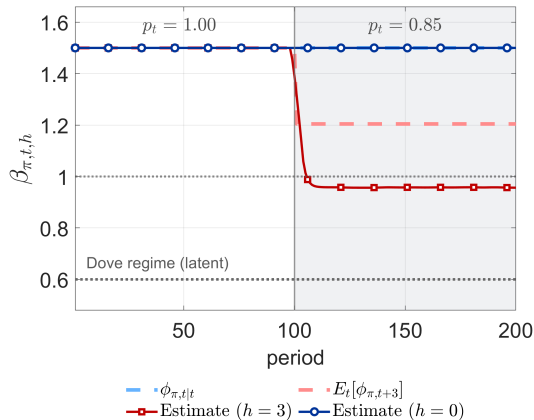
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- ✓ Nowcast ($h=0$): 1.50.

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3. $\mathbb{E}_t[\phi_{\pi,t+h}]$ vs. $\phi_{\pi,t|t}$: **distinct here, identical in the literature's practice**

- Constant rule, random walk, horizon-common coefficient: all equate them.

Roadmap

1. Simple Model of Perceived vs. Actual rule
2. Identification of Perceived Rule
3. **Empirical Estimation of Perceived Rule**
4. Subjective Perception Markov Switching New Keynesian Model.

Empirical Framework

1. Forward-looking Taylor rule specification:

$$\begin{cases} i_t = (1 - \rho_t) \left[\phi_{\pi,t} \pi_{t+1|t}^{GB} + \phi_{x,t} (-2 \cdot u_{t+1|t}^{GB}) \right] + \rho_{1,t} i_{t-1} + \rho_{2,t} i_{t-2} + c_t + \varepsilon_t \\ i_{t|t} = (1 - \rho_{t|t}) \left[\phi_{\pi,t|t} \pi_{t+1|t}^{BC/SPF} + \phi_{x,t|t} (-2 \cdot u_{t+1|t}^{BC/SPF}) \right] + \rho_{1,t|t} i_{t-1|t} + \rho_{2,t|t} i_{t-2|t} + \tilde{c}_t + e_t \\ i_{t+h|t} = (1 - \rho_{t,h}) \left[\beta_{\pi,t,h} \pi_{t+h+1|t}^{BC/SPF} + \beta_{x,t,h} (-2 \cdot u_{t+h+1|t}^{BC/SPF}) \right] + \rho_{1,t,h} i_{t+h-1|t} + \rho_{2,t,h} i_{t+h-2|t} + \tilde{c}_{t,h} + e_{t,h} \end{cases}$$

Use Stock and Watson (1998) and Boivin (2005) + Stochastic Volatility.

Empirical Framework

1. Forward-looking Taylor rule specification:

$$\begin{cases} i_t = (1 - \rho_t) \left[\phi_{\pi,t} \pi_{t+1|t}^{GB} + \phi_{x,t} (-2 \cdot u_{t+1|t}^{GB}) \right] + \rho_{1,t} i_{t-1} + \rho_{2,t} i_{t-2} + c_t + \varepsilon_t \\ i_{t|t} = (1 - \rho_{t|t}) \left[\phi_{\pi,t|t} \pi_{t+1|t}^{BC/SPF} + \phi_{x,t|t} (-2 \cdot u_{t+1|t}^{BC/SPF}) \right] + \rho_{1,t|t} i_{t-1|t} + \rho_{2,t|t} i_{t-2|t} + \tilde{c}_t + e_t \\ i_{t+h|t} = (1 - \rho_{t,h}) \left[\beta_{\pi,t,h} \pi_{t+h+1|t}^{BC/SPF} + \beta_{x,t,h} (-2 \cdot u_{t+h+1|t}^{BC/SPF}) \right] + \rho_{1,t,h} i_{t+h-1|t} + \rho_{2,t,h} i_{t+h-2|t} + \tilde{c}_{t,h} + e_{t,h} \end{cases}$$

Use Stock and Watson (1998) and Boivin (2005) + Stochastic Volatility.

2. Data:

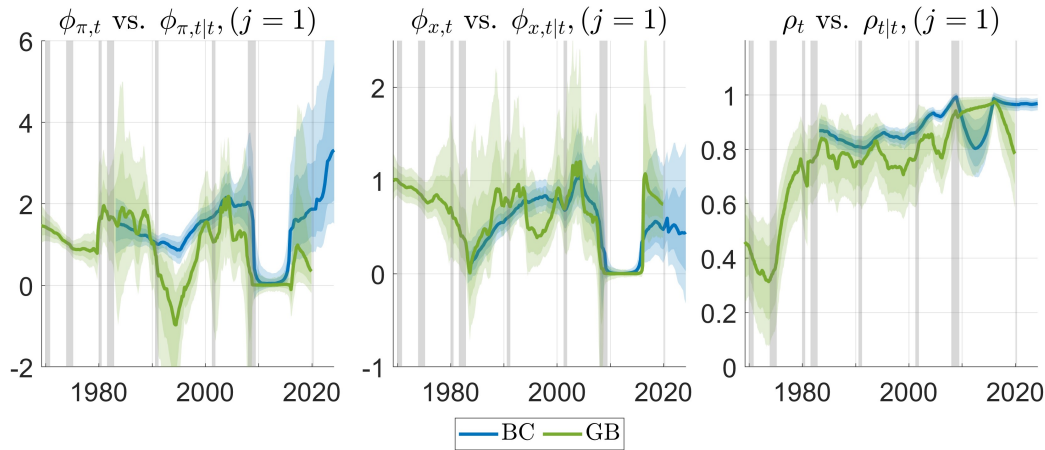
Actual Rule

- *Source:* Greenbook (Tealbook).
- *Sample:* 1969-2019, FOMC frequency.

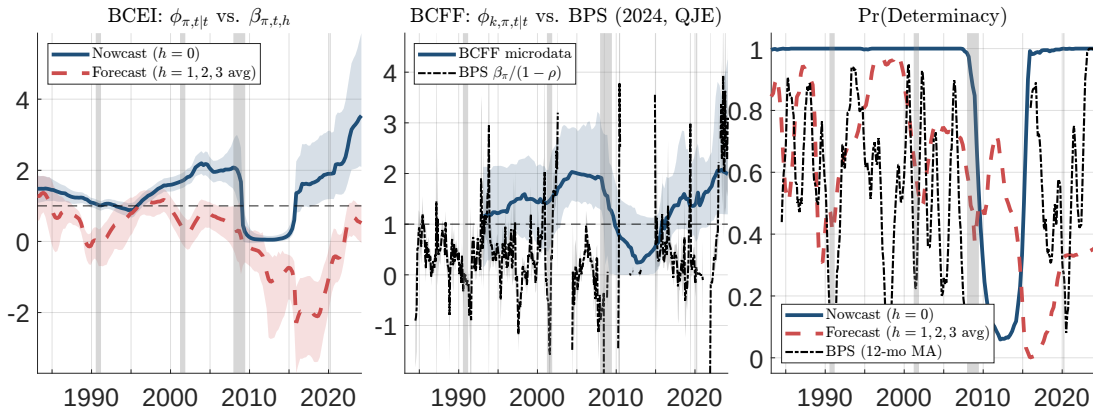
Perceived Rule

- *Baseline:* BCEI + SPF, 1983-2024, quarterly.
- *Robustness:* BCFF microdata.

Perceived vs. Actual Rule



Restoration of Perceived Taylor Principle



Roadmap

1. Simple Model of Perceived vs. Actual rule
2. Identification of Perceived Rule
3. Empirical Estimation of Perceived Rule
4. **Subjective Perception Markov Switching New Keynesian Model**

SP-MSNK Model: Framework and Solution

$$\left. \begin{array}{l} \text{Perceived } s_t^p \in \{1 \cdots S_p\} \text{ under subjective } \widehat{\mathbf{P}} \\ \text{Actual } s_t^f \in \{1 \cdots S_f\} \text{ under } \mathbf{P} \end{array} \right\} \Rightarrow \text{True state } \ell = (s_t^p, s_t^f)$$

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1. **Perceived Law of Motion.** Agents believe, and solve, given $s_t^p = j$:

$$\Gamma_{0,j}^P E_t^P[\mathbf{y}_{t+1}] + \Gamma_{1,j}^P \mathbf{y}_t + \Gamma_{2,j}^P \mathbf{y}_{t-1} + \Gamma_{3,j}^P \boldsymbol{\varepsilon}_t^p = \mathbf{0} \implies \mathbf{y}_{t|t} = \widehat{\mathbf{A}}_j^{\text{PLM}} \mathbf{y}_{t-1} + \widehat{\mathbf{B}}_j^{\text{PLM}} \boldsymbol{\varepsilon}_t^p$$

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2. **Expectations.** From that PLM, for $s_t^p = j$:

$$E_t^P[\mathbf{y}_{t+1}] = \sum_{k=1}^{S_p} \widehat{P}_{jk} \widehat{\mathbf{A}}_k^{\text{PLM}} \left(\widehat{\mathbf{A}}_j^{\text{PLM}} \mathbf{y}_{t-1} + \widehat{\mathbf{B}}_j^{\text{PLM}} \boldsymbol{\varepsilon}_t^p \right)$$

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3. **Actual Law of Motion.** Substituting $E_t^P[\cdot]$, the economy obeys, given ℓ :

$$\Gamma_{0,\ell}^A E_t^P[\mathbf{y}_{t+1}] + \Gamma_{1,\ell}^A \mathbf{y}_t + \Gamma_{2,\ell}^A \mathbf{y}_{t-1} + \Gamma_{3,\ell}^A \boldsymbol{\varepsilon}_t = \mathbf{0} \implies \mathbf{y}_t = \mathbf{A}_\ell^{\text{ALM}} \mathbf{y}_{t-1} + \mathbf{B}_\ell^{\text{ALM}} \boldsymbol{\varepsilon}_t$$

Implication 1 – Determinacy depends on perception

Model ($S_p = S_f = 1$)

$$x_t = E_t^P [x_{t+1}] - \tau \left(i_t - E_t^P [\pi_{t+1}] - r_t^n \right)$$

$$\pi_t = \beta E_t^P [\pi_{t+1}] + \kappa x_t$$

Rules

Perceived: $i_t = r_t^n + \phi_\pi^P \pi_t$

Actual: $i_t = r_t^n + \phi_\pi^F \pi_t$

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Under FIRE ($\phi_\pi^P = \phi_\pi^F$)

$$\pi_t = x_t = 0 \Rightarrow / \not\Leftarrow i_t = r_t^n$$

$$\Rightarrow / \Leftarrow i_t = r_t^n + \phi_\pi^F \pi_t$$

Peg (Galí 2015), then $\phi_\pi^F > 1$.

Determinacy is required to attain the target allocation $\pi_t = x_t = 0$.

Implication 1 – Determinacy depends on perception

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Peg (Galí 2015), then $\phi_\pi^F > 1$.

Dropping FIRE ($\phi_\pi^P \neq \phi_\pi^F$)

$$\pi_t = x_t = 0 \Rightarrow / \Leftarrow i_t = r_t^n$$

$$\text{Determinacy} \iff \phi_\pi^P > 1$$

Peg restored, without $\phi_\pi^F > 1$.

Determinacy is required to attain the target allocation $\pi_t = x_t = 0$.

Policy and empirical implications: Not $\phi_\pi^F > 1$, but $\phi_\pi^P > 1$.

SP-MSNK Model: Setup

IS and Phillips curve

$$x_t = E_t^P [x_{t+1}] - \tau (i_t - E_t^P [\pi_{t+1}]) + z_{Dt}$$

$$\pi_t = \beta E_t^P [\pi_{t+1}] + \kappa x_t + z_{St}$$

Rules ($S_p = S_f > 1$)

$$\text{Perceived: } i_t = \left(1 - \rho_{s_t^p}\right) \left[\phi_{\pi, s_t^p} E_t^P [\pi_{t+1}] + \phi_{x, s_t^p} E_t^P [x_{t+1}] \right] + \rho_{1, s_t^p} i_{t-1} + \rho_{2, s_t^p} i_{t-2}$$

$$\text{Actual: } i_t = \left(1 - \rho_{s_t^f}^F\right) \left[\phi_{\pi, s_t^f}^F E_t^P [\pi_{t+1}] + \phi_{x, s_t^f}^F E_t^P [x_{t+1}] \right] + \rho_{1, s_t^f}^F i_{t-1} + \rho_{2, s_t^f}^F i_{t-2} + \varepsilon_{Mt}$$

- s_t^p evolves under $\widehat{\mathbf{P}}$, s_t^f under $\mathbf{P}$, with $\mathcal{S}^p = \mathcal{S}^f = \mathcal{S} = \{1, \dots, S\}$.
- The Fed responds to E_t^P : to isolate the role of perceived vs. actual rules.

Counterfactual: Method

Given a sequence of $\mathbf{A}_\ell^{\text{ALM}}, \mathbf{B}_\ell^{\text{ALM}}$ for each t :

1. Recover structural shocks $\boldsymbol{\varepsilon}_t = [\varepsilon_{D,t}, \varepsilon_{S,t}, \varepsilon_{M,t}]'$ from data using baseline ALM:

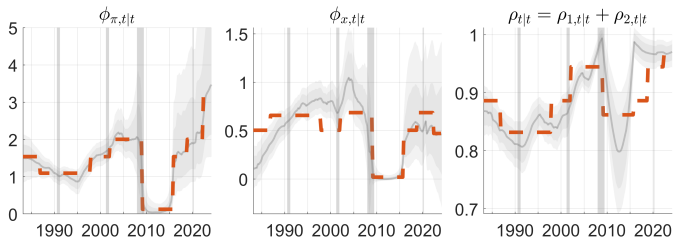
$$\mathbf{y}_t = \mathbf{A}_\ell^{\text{ALM}} \mathbf{y}_{t-1} + \mathbf{B}_\ell^{\text{ALM}} \boldsymbol{\varepsilon}_t$$

2. Construct and replace with counterfactual ALM: $(\mathbf{A}_{\ell^{\text{CF}}}^{\text{ALM}}, \mathbf{B}_{\ell^{\text{CF}}}^{\text{ALM}})$

t	$\dots$	T_1	$\dots$	T_2	$\dots$
$\ell = (s_t^p, s_t^f)$	$\dots$			(H, H)	$\dots$
ℓ^{CF1}	$\dots$			$(\mathbf{D}, H)$	$\dots$
ℓ^{CF2}	$\dots$		(H, H) but with different $\hat{P}$		$\dots$

3. Feed *same shocks* through counterfactual ALM $\Rightarrow$ counterfactual paths.

Calibration



- Univariate HMM on $\phi_{\pi,t|t}$, $K = 5$ by BIC $\Rightarrow (s_t^p)_{t=1}^T$.
- Other coefficients: within-regime averages.
- Baseline: $(s_t^p) = (s_t^f)$, giving ℓ .
- Dove (Lubik and Schorfheide 2004): unreachable in baseline, active in counterfactuals.

Regime-specific rule

	ϕ_{π}	ϕ_x	ρ	$\widehat{P}_{kk}$
1: Eagle	3.18	0.47	0.97	0.79
2: S. Hawk	2.00	0.69	0.94	0.86
3: Hawk	1.54	0.50	0.89	0.84
4: W. Hawk	1.09	0.66	0.83	0.87
5: ZLB	0.13	0.02	0.86	0.84
6: Dove	0.77	0.68	0.60	0.98

Structural

$$\beta = 0.99 \quad \tau = 0.50 \quad \kappa = 0.086$$

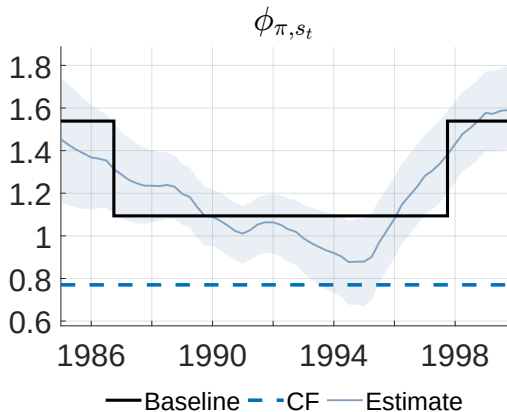
$$\rho_D = \rho_S = 0.85$$

ϕ_x annualized; $\rho = \rho_1 + \rho_2$ is total inertia.

Implication 2 – Source of the Great Moderation

Consensus: the Great Moderation followed a shift to a hawkish policy rule.

Clarida, Gali, and Gertler (2000), Coibion and Gorodnichenko (2011), and Bianchi (2013)

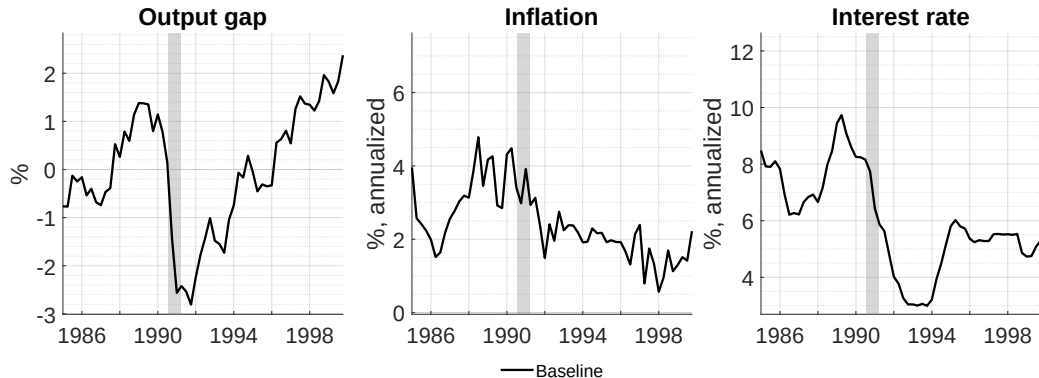


Baseline: Hawk **perception**, Hawk actual.

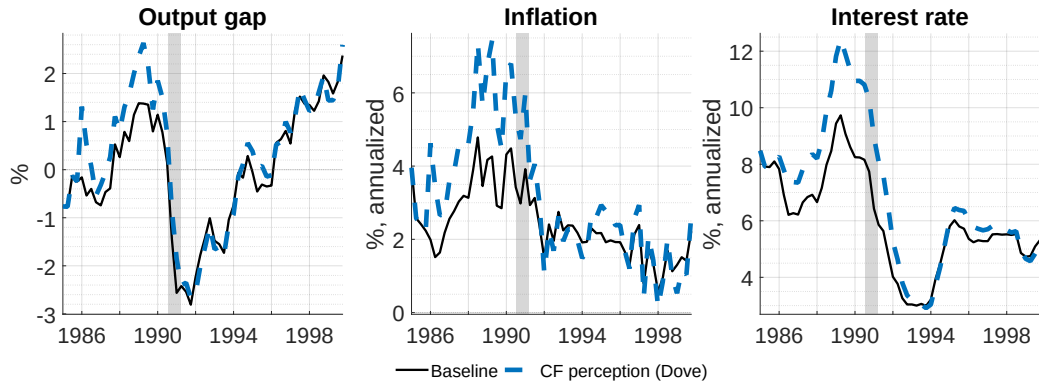
1. **Dove perception**, Hawk actual
 $\Rightarrow$ *Perception channel*
2. Hawk **perception**, **Dove** actual
 $\Rightarrow$ *Direct policy channel*

Q. What was the role of perception?

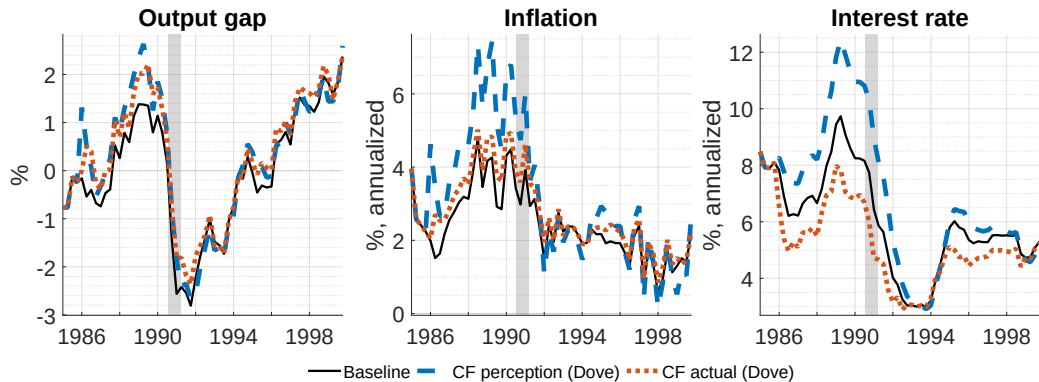
Implication 2 – Source of the Great Moderation



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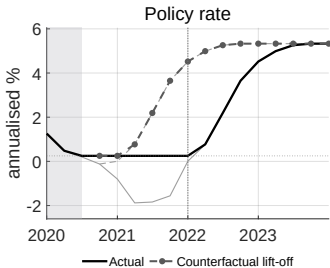
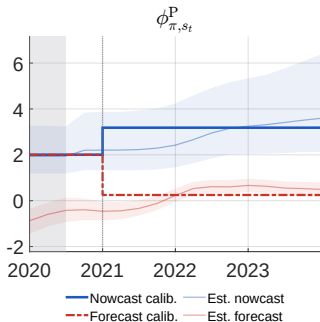


Implication 2 – Source of the Great Moderation



- Dovish beliefs amplify the level and volatility more than the Dovish rule.

Implication 3 – Misreading Perception (2021–2023)



Counterfactual: lift-off pulled at 2021Q1.

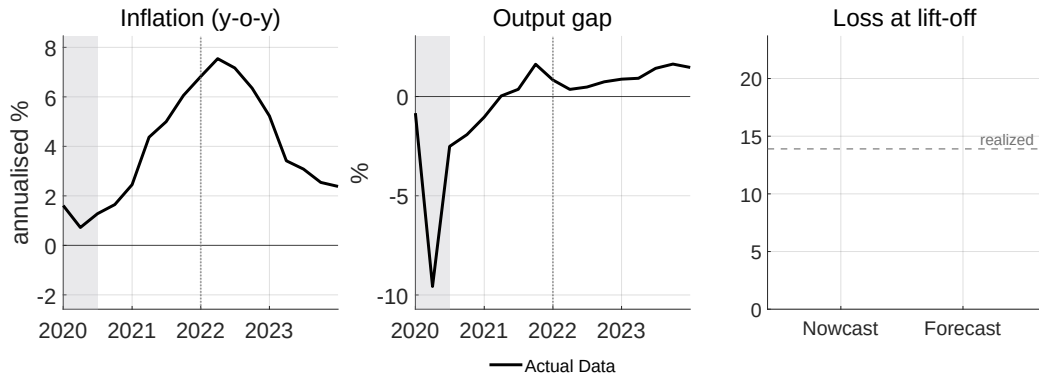
Two central banks read the *same* path:

1. **Nowcast**-calibrated ϕ_{π, s_t}^p
2. **Forecast**-calibrated ϕ_{π, s_t}^p

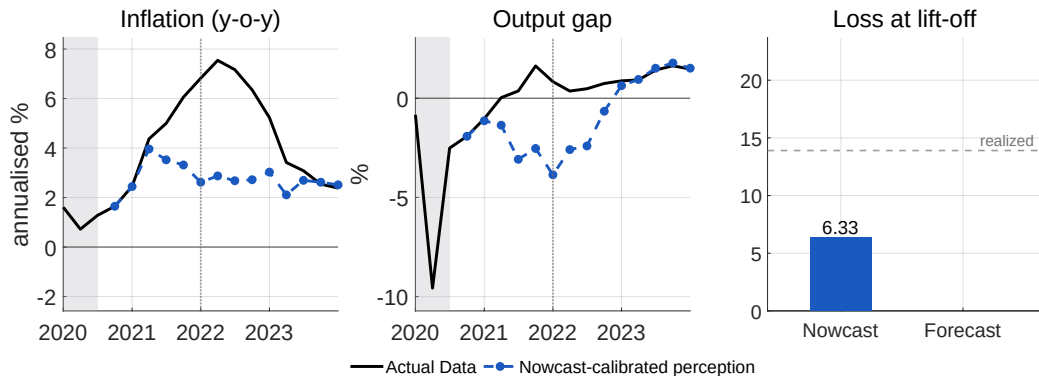
Each scores loss with:

$$\mathcal{L} = \mathbb{E} \left[\left(\pi_t^{\text{yoy}} - 2 \right)^2 + x_t^2 \right]$$

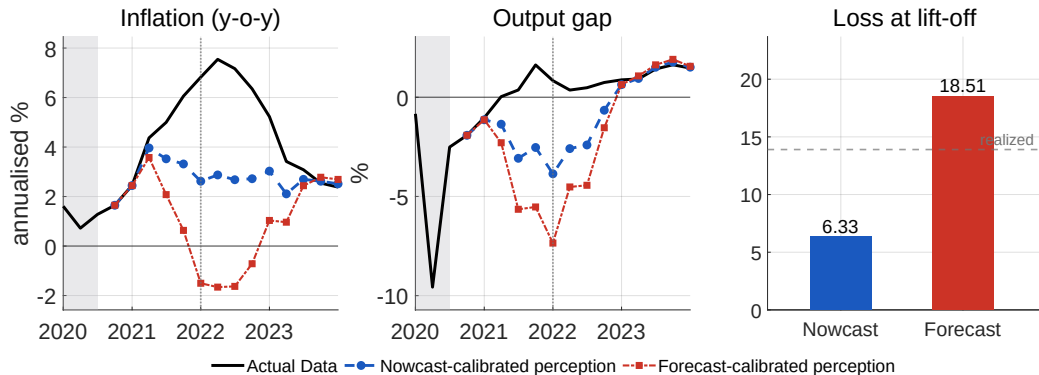
Implication 3 – Same Path, Different Readings



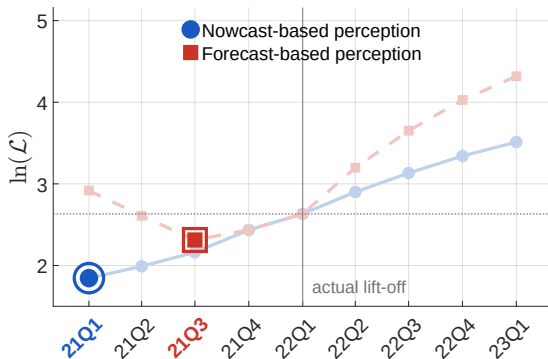
Implication 3 – Same Path, Different Readings



Implication 3 – Same Path, Different Readings



Implication 3 – The Cost of Misreading



- The same policy path is scored differently under the two **perceptions**.
- The **forecast**-based Fed rates the **nowcast**-optimal lift-off worse than its own optimal lift-off.
- Misreading **perception** leads to miscalculating policy outcome.

Conclusion and Contributions

1. Interest rate **forecasts** understate the **perceived** rule. **Nowcasts** recover it.
2. Determinacy depends on the **perceived** Taylor principle, not the actual one.
3. The Great Moderation required a shift in **perception**, not just policy.
4. Misreading **perception** may result in incorrect policy analysis.

*Reorienting the focus from the actual rule to the **perceived** rule.*

Appendix

Introduction

- ◀ Motivation
- ◀ Overview
- ◀ Literature
- ◀ Summary

Simple Model

- ◀ Setup
- ◀ PLM, ALM, Takeaways

Identification

- ◀ Setup, Covariance Bias
- ◀ Illustration
- ◀ Source and Sign of Bias
- ◀ Forecast vs. Nowcast
- ◀ Two-Regime Version

Empirics

- ◀ Framework
- ◀ Perceived vs. Actual
- ◀ Perceived Taylor Principle

SP-MSNK Model

- ◀ Model Setup
- ◀ PLM and ALM
- ◀ HMM Calibration
- ◀ Parameters

Counterfactuals

- ◀ Method
- ◀ Implication 2: Setup
- ◀ Implication 2: Results
- ◀ Implication 3: Setup
- ◀ Implication 3: Results

Determinacy

- ◀ Theory
- ◀ Proposition

Conclusion

- ◀ Conclusion

Summary

1. Identifying and estimating the **perceived** rule

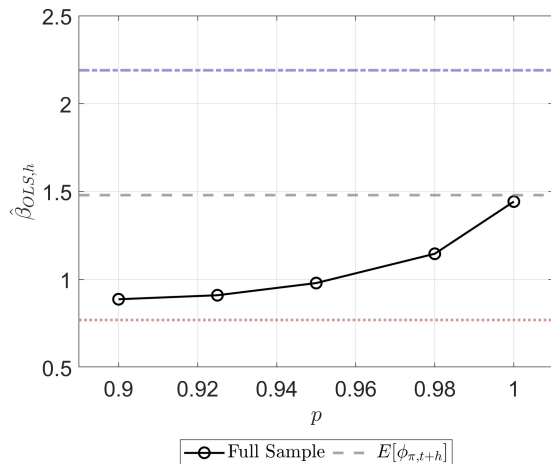
- *Bias*: Forecasts embed uncertainty about the future rule, biasing ϕ_π down.
- *Identification*: ✗ Interest rate forecast-based ✓ Nowcast-based.
- *Estimates*: ✗ Taylor principle under forecasts ✓ Restored under nowcasts.

2. Macroeconomic implications of the **perceived** rule (the SP-MSNK model)

- *Determinacy*: ✗ Implement the Taylor principle ✓ Anchor **perception** to it.
- *Great Moderation*: ✗ Shift in the policy regime ✓ Shift in **perception**.
- *Post-Covid liftoff*: ✗ Forecast-based - delayed lift-off ✓ Nowcast-based.

Reorienting the focus from the actual rule to the **perceived** rule.

Numerical Illustration - OLS Bias



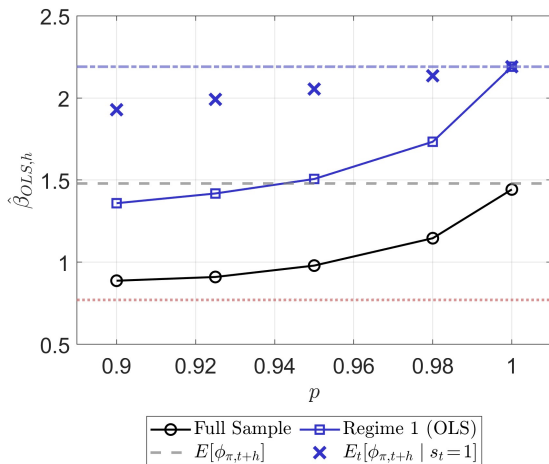
- **Unconditional OLS**

- Sample: Mixture of regimes
- Downward bias increases in regime uncertainty.
- Spurious Taylor principle violation.

- **Regime-Specific Subsample OLS**

- Sample: Regime 1 (Hawkish)
- Actual regime shift is not the precondition for bias.
- Subsample regression does not work.
- What matters is agent's uncertainty.

Numerical Illustration - OLS Bias



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SP-MSNK Model: Expectation Formation

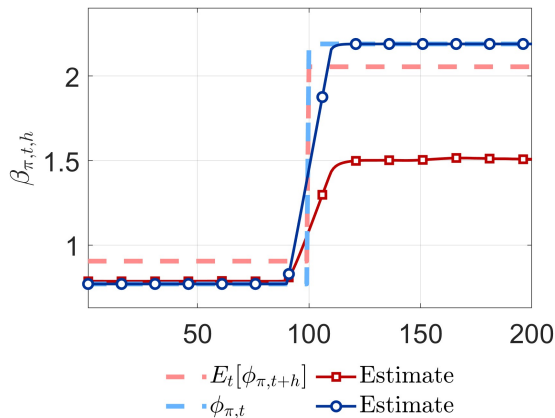
For some perceived state $s_t^p = j \in \{1 \cdots S_p\}$, Given PLM $\hat{\mathbf{A}}_j, \hat{\mathbf{B}}_j$ and $\hat{\mathbf{P}}$:

$$E_t^P [\mathbf{y}_{t+1}] = \sum_{k=1}^{S_p} \hat{P}_{jk} \hat{\mathbf{A}}_k \left(\hat{\mathbf{A}}_j \mathbf{y}_{t-1} + \hat{\mathbf{B}}_j \varepsilon_t^p \right)$$

Expectations depend on *all* regimes agents consider possible.

- $\hat{\mathbf{A}}_j, \hat{\mathbf{B}}_j$: Mainly depend on current perceived regime (s_t^p)
- Changing $\hat{\mathbf{P}}$ (e.g., credibility erosion) changes expectations.
 - Diagonal: Credibility of the current regime to persist
 - Off-diagonal: Belief in transitioning to other regimes.

Numerical Illustration - Two Regimes



Dovish for the first half of the sample, hawkish for the second.

1. Interest rate forecast ($h > 0$) leads to downward bias.
 - Subsample OLS / IV do not work.
 - Actual regime switch is not a precondition.
2. Use interest rate nowcast ($h = 0$) for unbiased perception.